Lauds Factories - Co-funded by the European Union, 2024-2026 GA 101135986

LAUDS FACTORIES

Open Call 2 | LAUDS Replication

Collaboration Report

DATA-RIDE

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Project coding LAUDSREP-01-19	*DATA-RIDE: Data and Advocacy for Transforming Active Riding & Infrastructure for a Democratic Environment*	
Deliverable title Production Process Blueprint: Data-Ride		**Due date** 01.10.2026
Hybrid Team entities **Lead partner:** re:edu GmbH & Co. KG **Partner 1: Laura Sobral** **Partner 2: Global Innovation Gathering (GIG)**		
Challenge Category Sustainable urban mobility	**Challenge Title** Citizen-led sensing and data-driven advocacy for safer cycling infrastructure	**Host LAUDS Factory** Université de Lorraine

1. Project Description

LAUDS Challenge

DATA-RIDE addressed the mobility challenge by connecting open-source technology, citizen participation, and advocacy. Developed in Nancy in collaboration with Université de Lorraine, the project explored how citizen-generated data could make everyday cycling conditions more visible and support evidence-informed discussions about urban mobility.
The project responded to three interconnected problems identified in its blueprint: everyday cycling experiences are underrepresented in conventional planning data; cycling safety depends on interactions between mobility modes as well as infrastructure; and local advocacy needs accessible evidence to support dialogue with institutions.

Using senseBox:bike devices adapted with a second ToF sensor, volunteers collected measurements during everyday journeys. Spatial analysis and participatory discussions transformed these recordings into shared insights, maps, communication materials, and policy recommendations. The resulting solution combines a technical sensing system with a community-based process for interpreting and communicating evidence.

Project Implementation

The initial proposal described a nine-month programme combining technical development, a three-week sensing campaign, visual analysis, participatory workshops, policy recommendations, and an advocacy toolkit.

The implemented campaign involved ten volunteers collecting data between February and May 2026. The devices recorded GPS position, speed, lateral distance, particulate matter, and motion-based indicators related to stopping and road-surface conditions.

The final report documents 193,328 observations across 357 trips, covering more than 1,000 kilometres and approximately 84 hours of valid recorded movement. Data were cleaned, processed, and linked to OpenStreetMap street segments and selected local datasets shared by the Métropole du Grand Nancy on cycling infrastructure, parking, and tree presence.

Initial objective	Outcome and adaptations
Extend the sensing system with dual-sided distance measurements	The deployed system supported lateral-distance and cycling-corridor analysis. Although the sensors performed successfully during the campaign, further calibration and refinement of the data-processing methods are still needed to improve data accuracy and reliability. Based on challenges identified during the deployment, local partners also supported the development and testing of an alternative mounting prototype for bicycles equipped with rear child seats.
Organise a three-week sensing campaign	Collection took place between February and May, extending a couple of weeks beyond the originally proposed three-week period because some participants were unable to join during the initially scheduled timeframe.
Produce maps and visual indicators	Spatial analyses were developed from sensor recordings and local contextual datasets. Maps were made available by local workshops and presented publicly at the Fête du Vélo 2026.
Co-create policy recommendations	Findings and local experiences were discussed during the Fête du Vélo and a workshop at the Lorraine Fab Living Lab on 1 June. These discussions guided the policy recommendation report.
Toolkit	The project developed communication resources and a toolkit documenting the relationship between sensing, interpretation, advocacy, and sustainability.

Disseminate results openly	Results were presented publicly and shared through project resources. The blueprint is licensed under CC BY 4.0.

The project achieved a locally implemented sensing campaign and a participatory interpretation process. However, the dataset remains exploratory rather than representative of all cycling conditions in Nancy. Overtaking detection also requires further local calibration and video-supported validation. The project produced recommendations, but their adoption by public authorities still has not been demonstrated.

From June to October 2026, activities focused on consolidating project outputs and preparing dissemination materials. This included preparing a blog post for the GIG network, organising the maps, methodology, Visual Toolkit and Policy Recommendation Paper, and completing the LAUDS Blueprint and Final Report. Documentation was reviewed and refined to clarify the findings, limitations and lessons learned, and materials were prepared for sharing through the project repository and LAUDS communication channels.

Communication and Dissemination

Communication formed part of the solution itself, helping participants translate measurements into accessible messages about shared responsibility in public space. Activities and outputs included:

- Initial workshop, mid-February: introduction to DATA-RIDE and senseBox:bike, device distribution, installation, and testing.
- Exploratory workshop, mid-February: Discussion with local cyclists to identify key topics and hypotheses to guide the data analysis and policy recommendations.
- Data collection campaign, February–May: Two rounds of five participants collected data over several weeks using the senseBox:bike equipped with the second ToF sensor.
- Fête du Vélo: public presentation and discussion of the project and emerging results.
- Participatory workshop at Lorraine Fab Living Lab, June : discussion involving cyclists, researchers, local organisations, and mobility stakeholders.
- Maps and accompanying presentation: spatial visualisations connecting findings with streets, corridors, and urban conditions in Nancy.
- Policy recommendation report: documentation of measured findings, participant experiences, and proposed directions for action.
- Communication materials: Visual materials informed by collective discussions, using the collected data as a basis.
- Toolkit: resources explaining the overall process and supporting adaptation by other communities.

The report explicitly identifies DATA-RIDE as developed within the LAUDS Factories programme in collaboration with Université de Lorraine.

Open reuse is supported through documentation of the hardware, software, data workflow, workshop formats, and communication outputs. The blueprint carries a CC BY 4.0 licence. Other outputs require their respective licences and clear access conditions, particularly for location data.

Team and skill development

Member/entity	Responsibilities and expertise
re:edu	Technical preparation of the senseBox:bike with a second ToF, adaptation of the application and implementation of the French Version - expertise on open hardware, open data platform and software development.
Artur Kim Shum	Coordination, spatial analysis, participatory sensing, and visual communication - expertise on urban planning, and innovation design.
Laura Sobral	Participatory methods, workshop facilitation, collective interpretation, and advocacy - experience in participatory urbanism and collaborative governance.
Global Innovation Gathering (GIG)	Administration, dissemination, and transferability: External communication and knowledge sharing through international open-technology networks, reaching a worldwide community and supporting the transferability of the project.
Université de Lorraine / host team	Local research collaboration, stakeholder engagement, workshop support, laboratory prototyping, volunteer coordination, local communication, and connection with local policymakers. - Anaelle Hily and Laurent Dupont

Collaborative work inside the hybrid team

The team worked across technical, participatory, and policy-oriented activities. Technical work focused on developing the sensing system and generating, processing, and analysing the

collected data, while the creative and facilitation work helped identify relevant local questions, interpret the findings, and communicate the results.

At the beginning of the project, the hybrid team held regular online meetings to share updates on technical development and coordinate local engagement. The Nancy team played an important bridging role by connecting the project with local stakeholders, cycling organisations, and participants. During the field visits in February and May, the teams jointly facilitated workshops and communicated with participants and local organisations working on cycling and urban mobility.

Throughout the data collection campaign, the Nancy team maintained direct contact with the participants. Challenges encountered during deployment were brought back to the regular team meetings, where possible solutions and adaptations were discussed collectively to allow the campaign to continue.

Decision-making combined different areas of expertise. Technical decisions were primarily led by the re:edu team, while Laura took a stronger role in the policy-related work, always in consultation with the Nancy partners. Participants contributed knowledge about routes, unsafe locations, overtaking conditions, visibility, and infrastructure gaps, while researchers and local actors provided technical and institutional context. All major reports and project outputs were reviewed collectively by the team and can therefore be considered co-authored outputs.

2. Collaboration with LAUDS Factories

Collaborative Experience with your host

Collaboration with Université de Lorraine and the Lorraine Fab Living Lab connected DATA-RIDE with Nancy's research and cycling ecosystem. It supported a process in which sensor evidence could be discussed alongside local experiences and institutional social participation knowledge.

The Lorraine Fab Living Lab hosted the three workshops in February and June, bringing together cyclists, researchers, local organisations, and mobility stakeholders. The place and institution mobilized engaged volunteers and participants for collective interpretation and discussion of possible responses.

Local laboratory support also contributed to development and printing a mounting prototype, addressing a practical aspect of deploying sensors on bicycles with racks.

Visits and Events

Visit/event	Time	Contribution
Initial workshop and Hypotheses meeting	(3 days) 11-13th February 2026	Introduction to the project, installation and testing of the devices, preparation of volunteers for the data collection campaign, and discussion of the main cycling issues identified by participants.
Online	March / April	Participation in two online workshops with students from Université de Lorraine.
Fête du Vélo	30 Mai 2026	Public presentation and discussion of results.
Lorraine Fab Living Lab workshop	(7 days) May 28th-June 3rd - 2026	Collective interpretation and development of recommendations and communication directions.
Final LAUDS event	October 2026	Exchange and dissemination within the LAUDS network represented by the Nancy Partners

Relation to the researchers and coaches

Interactions with the LAUDS coaches took place mainly online during the coaching sessions, with some relevant exchanges focusing on topics closely related to the project, particularly during the mid-term meeting. Towards the end of the project, other LAUDS teams also reached out to exchange experiences and lessons learned regarding fabrication and project implementation.

3. Collaboration with other actors

DATA-RIDE connected participating cyclists with researchers, local organisations, and mobility stakeholders through the local team at the University of Lorraine and the Lorraine Smart Cities Living Lab.

These actors contributed at different stages of the project. From the beginning, local organisations showed strong interest in the technology, leading to valuable discussions during the

first workshop introducing the senseBox:bike. Several organisations represented in this session also contributed volunteers to the data collection campaign and participated in discussions around hypotheses and key questions concerning cycling conditions in Nancy.

Beyond civil society organisations, representatives from the Métropole du Grand Nancy working on urban mobility participated in project discussions and provided local datasets on cycling infrastructure, parking and trees. These datasets supported contextual analysis of the sensor observations. Their contribution involved discussion and data provision; no formal municipal review or endorsement of the final maps or Policy Recommendation Paper is documented.

The local network also enabled DATA-RIDE to participate in the Fête du Vélo, a major cycling event in the region, held in a central public square in Nancy. The event provided an opportunity to present the project and its maps to a larger and more diverse audience, discuss the findings with local cyclists and residents, and gather additional feedback on the results.

The local ecosystem included the Métropole du Grand Nancy, Lorraine Smart Cities Living Lab, EDEN, Atelier Dynamo, Laboratoire Sauvage, volunteer cyclists, and workshop participants. Their contributions ranged from providing local knowledge and data to recruiting volunteers, hosting workshops and presentations, supporting the data collection campaign, and facilitating local outreach.

Participant numbers were estimated from photographic records. The Nancy partner holds attendance lists for some workshops, which may support further verification. These figures represent attendance per activity, not unique individuals; some people may have participated in more than one event.

- **February: Initial workshop — 17 participants:** Project introduction and device installation with local cyclists from EDEN, Atelier Dynamo and Laboratoire Sauvage, and FabLab users.
- **February: Local cycling hypotheses workshop — 10 participants:** Local cyclists connected to the Université de Lorraine Living Lab discussed cycling conditions and potential improvements.
- **February: Technical meeting with Leonel, Nancy mobility representative:** The project was presented and access to local datasets was requested. These datasets were subsequently shared. Attendance was not recorded.

- **Abril : Two university workshops — approximately 20 students in total:** Université de Lorraine students participated through the course “Transformations numériques au service des villes et des citoyens” and an innovation workshop.
- **May/June: Fête du Vélo — approximately 45 participants:** The project and data were presented in a public square to people from diverse backgrounds, including cyclists and other visitors interested in the findings.
- **May/June: data and policy brief workshop — 5 participants:** Representatives from EDEN, Atelier Dynamo and Laboratoire Sauvage discussed the findings and policy recommendations.

Feedback and Recommendations

Our experience with the LAUDS Open Call 2 was valuable in enabling DATA-RIDE to bring together open-source technology, participatory methods and local advocacy within an experimental project, while collaboratively advancing the development and adaptation of the senseBox device. Collaboration with the Université de Lorraine and the local ecosystem was important in connecting technical development with community engagement and collective learning.
At the programme level, however, we encountered several challenges that we believe are relevant to improving the support provided to future experimental and participatory initiatives.

Clarity, continuity and predictability of support
Project milestones, deadlines and expectations regarding deliverables were not always communicated sufficiently in advance. Some communications and feedback arrived later than expected, while changes in the people attending meetings occasionally affected continuity and required the team to revisit previously discussed matters. Although individual exchanges were generally constructive, the overall coordination process, largely organised around milestones and scheduled meetings, could have benefited from greater consistency, clearer timelines and more proactive follow-up.

Proportionality of requirements and the nature of the support
The administrative and reporting requirements, particularly during the final review, sometimes appeared disproportionate to the scale, resources and experimental scope of the project. The process increasingly felt oriented towards formal assessment and approval of deliverables, rather than providing ongoing support to a team testing an innovative approach under real-world conditions, with citizens actively contributing to its development and outcomes.
We recognise the importance of accountability and transparent documentation. Nevertheless, the level of formalisation expected should remain proportionate to the funding, duration, and prototypical and participatory nature of the projects being supported. A more collaborative review process, focused on identifying feasible improvements and recognising achievements and limitations, would have been particularly valuable.

Recognition of participatory processes and evolving outcomes

DATA-RIDE was conceived as a participatory and experimental initiative. While its original objectives remained central, specific activities, formats and outputs evolved through implementation, collective reflection, emerging needs and practical constraints.
During the final review, we felt that the emphasis on correspondence with the originally described outputs did not sufficiently recognise the processes through which they had been developed, the reasons for their adaptation or their practical relevance to the intended audiences. The replication manual provides an example. Following earlier indications that no specific format was prescribed, the team developed complementary resources, a Visual Toolkit and a Policy Recommendation Paper, supported by technical documentation, addressing different audiences and uses. Subsequent requests introduced more specific expectations, including formal validation protocols and communication templates, which had not been explicitly established as such in the approved proposal.
We believe that participatory projects should be evaluated not only against their initial descriptions but also in relation to the quality of their processes, the relevance of the adaptations made, the learning generated and the extent to which their original objectives have been achieved.

Alignment between evaluation criteria and project roles
Some requirements raised during the final assessment appeared closer to those typically associated with academic research, both in their format and substantive content, particularly concerning formal methodological validation.
While technical reliability and scientific rigour are important, our team's contribution was primarily focused on experimental implementation, community participation, collective interpretation and the development of accessible advocacy resources. The production of academic research standards and the fulfilment of formal academic validation requirements were not part of our specific responsibilities within the project.
Evaluation expectations should reflect the different responsibilities of academic, technical and creative or community-oriented partners. Additional validation or research requirements should be clearly established at the outset, with appropriate resources and responsibilities assigned.

Realistic expectations of impact, transferability and knowledge translation
Experimental projects can establish methods, generate exploratory evidence, develop practical resources and initiate relationships that enable future action. However, the subsequent adoption of recommendations, formal scientific validation or implementation in other contexts may depend on external actors, additional resources and longer timeframes.
Moreover, we question the applicability of the term replication to the kind of context-sensitive, participatory process developed through DATA-RIDE. Rather than a model to be technically reproduced in another location, we understand the project's value as providing inspiration, adaptable methodologies, practical knowledge and lessons that other communities can interpret and apply according to their own circumstances.
In this sense, knowledge translation, methodological adaptation or transferability may be more appropriate concepts than replication, particularly when the process itself depends on local relationships, priorities, institutional conditions and collective decision-making.
Distinguishing immediate project outputs from longer-term impacts, and recognising the contextual nature of their potential transfer, would support more realistic assessments of what small-scale initiatives can reasonably achieve within their funded period.

Recommendations for future open calls

Based on this experience, we recommend clearer and earlier communication of deliverable requirements and project expectations, a clearer roadmap of activities and deliverables, greater continuity in programme coordination, more proportionate administrative and evaluation demands, and stronger recognition of the evolving nature of participatory experimentation.

We also encourage greater attention to the specific characteristics, roles and contributions of individual projects, even when they occupy the same formal position within an open call or are subject to a common reporting framework.

The online training and capacity-building programme was extensive, but some sessions were less relevant to our project because their content did not correspond to its stage of development or particular needs. An overview of the programme and greater flexibility in matching training opportunities to individual projects could improve its usefulness.

Above all, we encourage LAUDS to reflect on how its meetings, reviews and support mechanisms are structured and experienced by participating teams. Rather than being perceived primarily as moments of evaluation, compliance and approval, these interactions could more deliberately foster mutual learning, constructive dialogue and recognition of the diverse forms of value generated by experimental projects.

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